Deployment & Operations Guide

Data Vault 2.0 Tax System Data Warehouse

Google Cloud VM Deployment • Performance Benchmarks

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PHASE A: GOOGLE CLOUD VM DEPLOYMENT

1. Overview and Prerequisites

This phase creates a clean Google Cloud VM environment for deploying the complete Data Vault 2.0 data warehouse. Deploying to a cloud VM first ensures that all benchmarks (Phase B) are captured on consistent, documented hardware — critical for reproducible results in your thesis.

1.1 Why Deploy to Cloud VM First?

- Consistent hardware specs for benchmark reproducibility (Table 5.7 + 5.8)
- Clean environment eliminates local machine interference (antivirus, background processes)
- Shareable with advisor via RDP for live demonstration
- Production-like deployment demonstrates real-world readiness

1.2 Prerequisites

- Google Cloud account with Free Trial ($300 credit for 90 days)
- All 9 SQL scripts (01-09) and SSIS packages ready on your local machine
- RDP client (built into Windows, or Microsoft Remote Desktop for Mac)

Cost Estimate: e2-standard-4 with Windows Server 2025: ~$0.38/hour. SQL Server 2025 Developer Edition is installed manually (free license). Google Cloud Free Trial ($300) covers the entire benchmark period. Always stop VM from Console when not in use.

2. Create Google Cloud VM

Step 2.1: Navigate to Google Cloud Console

1. Log in to https://console.cloud.google.com with your Google account.
2. Activate Free Trial if prompted ($300 credit for 90 days).
3. Go to Compute Engine > VM instances (enable Compute Engine API if prompted).
4. Click 'Create instance'.

Step 2.2: Configure Machine

Setting	Actual Value
Name	dv2-tax-system
Region / Zone	asia-southeast1 (Singapore) / asia-southeast1-a
Machine series	E2 (General purpose, low cost)
Machine type	e2-standard-4 (4 vCPUs, 2 cores, 16 GB RAM)
Provisioning model	Standard

Step 2.3: Configure Boot Disk (OS and Storage)

5. Click 'Change' under Boot disk.

Setting	Actual Value
Operating system	Windows Server
Version	Windows Server 2025 Datacenter (Server with Desktop Experience)
Boot disk type	SSD persistent disk
Size	128 GB

6. Click 'Select' to confirm boot disk settings.

Step 2.4: Networking

Leave defaults. RDP firewall rule (default-allow-rdp, TCP 3389) is already pre-configured in Google Cloud default VPC network.

Step 2.5: Create

7. Click 'Create'. Wait ~2-3 minutes for deployment.
8. VM appears in the instances list with a green checkmark and External IP.

Screenshot: Capture the VM instances page — documents your benchmark environment.

3. Connect and Configure VM

Step 3.1: Set Windows Password

9. In Google Cloud Console > VM instances > Click 'dv2-tax-system'.
10. Click 'Set Windows password'. Enter username (e.g., sotso). Click 'Set'.
11. Copy the generated password immediately — it is only shown once.

Step 3.2: RDP Connection

12. Click the RDP dropdown > Download the RDP file.
13. Open the .rdp file (or use mstsc.exe with the VM's External IP).
14. Enter the username and password. Accept the certificate warning.

Tip: For better RDP performance from Cambodia, Singapore region provides ~30ms latency vs ~200ms for US regions.

Step 3.3: Verify VM is Ready

15. Open PowerShell as Administrator.
16. Run: systeminfo | findstr /B /C:"OS Name" — confirm Windows Server 2025 Datacenter.
17. Verify F: drive is accessible and folders exist.

Note: SQL Server is NOT pre-installed. You will install it in the next section.

4. Install SQL Server and Development Tools

This section installs all required software on the clean Windows Server VM. Follow the steps in order.

Step 4.1: Download SQL Server 2025 Developer Edition

18. Open Microsoft Edge (pre-installed) on the VM.
19. Go to: https://www.microsoft.com/en-us/sql-server/sql-server-downloads
20. Scroll down to 'Developer' edition > Click 'Download now' (free, full-featured).
21. Run the downloaded installer. Select 'Custom' installation type.
22. Choose download location: C:\SQL2025\. Click Install to download media (~1.3 GB).

Step 4.2: Install SQL Server 2025

23. After download completes, the SQL Server Installation Center opens.
24. Click 'Installation' (left panel) > 'New SQL Server stand-alone installation'.
25. Accept license terms. Click Next through Product Updates and Install Rules.

Feature Selection

26. Check the following features:

- Database Engine Services (required)
- Integration Services (SSIS runtime — required for dtexec)
- Client Tools Connectivity

Instance Configuration

27. Select 'Default instance' (MSSQLSERVER). Click Next.

Server Configuration

28. SQL Server Database Engine service: Automatic startup.
29. SQL Server Agent: Automatic startup (for scheduled jobs).

Database Engine Configuration

Setting	Value
Authentication Mode	Mixed Mode (Windows + SQL Authentication)
SA Password	Set a strong password (needed for remote SSMS access)
Add Current User	Click 'Add Current User' to make your Windows account admin
Data root directory	C:\Program Files\Microsoft SQL Server\ (default)
User database directory	C:\Program Files\Microsoft SQL Server\...\Data (default)

User log directory	C:\Program Files\Microsoft SQL Server\...\Data (default)
Backup directory	C:\Program Files\Microsoft SQL Server\...\Backup (default)
TempDB data directory	C:\Program Files\Microsoft SQL Server\...\Data (default)

Note: All data files reside on the 128 GB SSD boot disk. For this benchmark environment, the single SSD provides consistent I/O performance.

30. Click Next > Click Install. Installation takes approximately 10-15 minutes.
31. Verify the completion screen shows all features succeeded.

Result: SQL Server 2025 Developer Edition is now installed with SSIS runtime and Mixed Mode authentication.

Step 4.3: Install SQL Server Management Studio (SSMS)

32. Open Edge > Go to: https://aka.ms/ssmsfull
33. Download the latest SSMS installer (SSMS-Setup-ENU.exe, ~900 MB).
34. Run the installer. Click Install (~5-10 minutes).
35. Restart the VM if prompted.

Verify SQL Server + SSMS

36. Open SSMS from the Start menu.
37. Connect to Server: localhost, Windows Authentication.
38. Run: SELECT @@VERSION — confirm SQL Server 2025 Developer Edition (RTM 17.0.1000.7).
39. Run: SELECT SERVERPROPERTY('InstanceDefaultDataPath') — confirm data path.

Verify: If @@VERSION returns 'Microsoft SQL Server 2025 (RTM) - 17.0.1000.7', SQL Server is correctly installed.

Step 4.4: Configure SQL Server Memory

Limit SQL Server memory to leave room for Windows, VS, and SSIS:

```
-- Run in SSMS:
EXEC sp_configure 'show advanced options', 1; RECONFIGURE;
EXEC sp_configure 'max server memory', 12288; RECONFIGURE;  -- 12 GB of 16 GB
```

Step 4.5: Install Visual Studio 2026 Community

40. Download Visual Studio 2026 Community from https://visualstudio.microsoft.com/downloads/
41. In Workloads, select 'Data storage and processing'. Click Install (~15 min).

Step 4.6: Install SSIS Extension

42. Open Visual Studio > Extensions > Manage Extensions.
43. Search 'SQL Server Integration Services Projects'. Install and restart VS.

5. Deploy Database and ETL

Step 5.1: Transfer Files to VM

Option A (Google Drive): Upload scripts/SSIS packages to Google Drive from local machine. Sign into Google Drive on the VM's browser and download to C:\DV2_TaxSystem\.

Option B (RDP Clipboard): In RDP client, Options > Local Resources > More > Check 'Drives'. Your local drives appear in the VM. Copy project to C:\DV2_TaxSystem\.

Step 5.2: Execute SQL Scripts (in order)

#	Script	Purpose	Time
1	01_CreateDatabaseStructure.sql	Source DB + tables	~5 sec
2	02_TransactionData.sql	Generate ~60K records	~30 sec
3	03_ETL_Control_Setup.sql	ETL Control DB + procedures	~5 sec
4	04_DDL_Architecture_DW.sql	DW schemas (Staging/Bronze/Silver/Gold)	~10 sec
5	05_ETL_Staging_Procedures.sql	9 staging procedures	~5 sec
6	06_ETL_Bronze_Procedures.sql	23 Bronze procedures (Hub/SAT/Link)	~10 sec
7	07_ETL_Silver_Procedures.sql	7 Silver procedures (PIT/BRG/BUS)	~5 sec
8	08_ETL_Gold_Procedures.sql	11 Gold procedures (DIM/FACT)	~5 sec
9	09_ETL_Master_Orchestration.sql	Master pipeline procedure	~5 sec

Verification: After each script, check Messages tab. All should complete with 'Commands completed successfully.'

Step 5.3: Deploy SSIS Project

44. Open the SSIS solution (.sln) in Visual Studio on the VM.
45. Update all 7 Connection Managers to point to localhost.
46. Build solution (Ctrl+Shift+B) — verify 0 errors.
47. Test: Run Master_Complete_Pipeline.dtsx (F5). Wait for all green checkmarks.

Connection Strings: Update all 7 CMs: CM_TaxSystemDB, CM_DV_Staging, CM_DV_Bronze, CM_DV_Silver, CM_DV_Gold, CM_ETL_Control, CM_ETL_Control_OnError — all must use localhost.

6. Share with Advisor

Option A: RDP Access (Recommended)

48. Google Cloud Console > VPC Network > Firewall > Create rule: Allow TCP 3389 from advisor's IP.
49. Share: External IP, username, password with Mr. Chap Chanpiseth.
50. Advisor connects via Remote Desktop — full access to SSMS, VS, and all databases.

Option B: Remote SSMS Connection

51. Create firewall rule for port 1433 restricted to advisor's IP.
52. Share: IP, port 1433, SQL credentials. Advisor connects from their own SSMS.

Option C: Database Backup Files

```
BACKUP DATABASE TaxSystemDB TO DISK = 'C:\SQLBackup\TaxSystemDB.bak' WITH COMPRESSION;
BACKUP DATABASE DV_Staging TO DISK = 'C:\SQLBackup\DV_Staging.bak' WITH COMPRESSION;
BACKUP DATABASE DV_Bronze TO DISK = 'C:\SQLBackup\DV_Bronze.bak' WITH COMPRESSION;
BACKUP DATABASE DV_Silver TO DISK = 'C:\SQLBackup\DV_Silver.bak' WITH COMPRESSION;
BACKUP DATABASE DV_Gold TO DISK = 'C:\SQLBackup\DV_Gold.bak' WITH COMPRESSION;
BACKUP DATABASE ETL_Control TO DISK = 'C:\SQLBackup\ETL_Control.bak' WITH COMPRESSION;
```

Upload .bak files to Google Drive and share the download link.

7. Cost Management

7.1 Stop VM When Not in Use

Critical: Always stop from Google Cloud Console (Compute Engine > VM instances > Stop). This deallocates compute resources. You only pay for the disk when stopped.

7.2 Auto-Shutdown

53. Google Cloud Console > VM instance > Edit > Set a time limit for the VM, or use Instance Schedules.
54. Alternative: Use Cloud Scheduler or a simple cron job to auto-stop.

7.3 Cost Summary

Resource	Cost/Hour	Monthly (8h/day)
e2-standard-4 Compute	~$0.17/hr	~$121
Windows Server License	Included	~$134
Boot Disk (128 GB SSD)	Fixed	~$24
SQL Server License (Developer)	Free	$0
TOTAL	**~$0.38/hr**	**~$279/month**

Google Cloud Free Trial: $300 credit for 90 days. For benchmark usage (~3-5 hours total), actual cost is only $1-2. Stop VM when not in use.

7.4 Cleanup After Project

55. Google Cloud Console > Compute Engine > VM instances > Select dv2-tax-system > Delete.
56. Also delete the boot disk when prompted.

Warning: Deletion is permanent. Export all benchmark results, screenshots, and backups before deleting.

PHASE B: PERFORMANCE BENCHMARKS

Run all benchmarks on the Google Cloud VM for consistent, reproducible results that match the hardware documented in Table 5.7/5.8 of the final report.

8. Benchmark Overview and Script Files

8.1 Report Table Mapping

Report Table	Benchmark Description	Script Section	Captures
Table 5.7+5.8	Hardware & Software Specification	[0] Environment	CPU, RAM, OS, SQL
Table 5.1	ETL Load Time (Full vs Incremental by Layer)	[1] ETL Times	Layer-by-layer sec
Table 5.2	Query Performance (Gold vs Raw Vault)	[2] Query Perf	Simple/Med/Complex ms
Table 5.3	Scalability Analysis	Section 12	Multi-scale throughput

8.2 Script Files Required

File	Purpose
00_CleanAll_FreshFullLoad.sql	Reset all databases to empty state
10_Verify_FullLoad.sql	Verify full load row counts and integrity
11_Verify_IncrementalLoad.sql	Verify incremental load with delta comparison
12_IncrementalTest_SourceChanges.sql	Insert/Update test data for incremental testing
13_Performance_Benchmarks.sql	Comprehensive benchmark suite (ETL, query, storage)

9. Clean Environment + Full Load

Step 9.1: Clean All Databases

57. In SSMS on the Google Cloud VM, open and execute 00_CleanAll_FreshFullLoad.sql.
58. Verify all 15 rows show Rows = 0 in the Results grid.

Figure 5.5: Clean Environment Verification — All Data Warehouse Tables Reset to Zero Rows Before Benchmark

Step 9.2: Execute Full Load via SSIS

59. In Visual Studio, open Master_Complete_Pipeline.dtsx.
60. Press F5 (Start Debugging). Wait for all green checkmarks (~60-75 seconds).
61. Note the execution time from the Progress tab.
62. Click Stop Debugging (Shift+F5).

Figure 5.6: SSIS Master Pipeline Full Load Execution — All 49 Steps Completed Successfully

Step 9.3: Verify Full Load

63. In SSMS, open and execute 10_Verify_FullLoad.sql.
64. Review row counts: STG_Taxpayer=1,000 | STG_MonthlyDeclaration=38,461 | FACT_MonthlyDeclaration=38,461.
65. Check Summary Verdict in the Messages tab.

Expected: 49 tables across 4 databases, all populated. Verify all SATs, Hubs, Links, DIMs, and FACTs have non-zero row counts.

Figure 5.7: Full Load Row Count Verification — 49 Tables Across 4 Databases with Expected Counts

10. Incremental Load Test

Step 10.1: Insert Test Data Changes

66. In SSMS, open and execute 12_IncrementalTest_SourceChanges.sql.
67. Verify: 16 updates (Taxpayer, Officer, Category, Owner, Declaration, Payment) + 13 inserts (new Taxpayer, Owner, Declaration, Payment, AnnualDecl).
68. Verify source counts: Taxpayer=1,002, MonthlyDeclaration=38,465.

Important: The script sets UpdatedDate = GETDATE() on all updated records, enabling watermark-based incremental detection.

Step 10.2: Execute Incremental Load via SSIS

69. In Visual Studio, press F5 on Master_Complete_Pipeline.dtsx.
70. Observe: EXT_EvaluateLoadType routes to EPT_Master_Incremental_Load (not Full Load).
71. Wait for all green checkmarks (~50-65 seconds).

Figure 5.8: SSIS Master Pipeline Incremental Load Execution — Watermark-Based Delta Processing Path

Step 10.3: Verify Incremental Load

72. In SSMS, open and execute 11_Verify_IncrementalLoad.sql.
73. Check Step Log Delta: HUB_Taxpayer +2, SAT_Taxpayer +5 (3 updated + 2 new), SAT_Officer +2.
74. Check SAT versioning: HistoricRows > 0 for modified entities.
75. Check DIM SCD2: ExpiredRows > 0 (DIM_Taxpayer, DIM_Officer).

Figure 5.9: Incremental Load Change Detection — ETL Step Log with Delta Records and SAT Versioning Metrics

11. Performance Benchmark Execution

Step 11.1: Run Benchmark Script

76. In SSMS, open 13_Performance_Benchmarks.sql.
77. Set Results to Grid mode (Ctrl+D).
78. Execute the entire script (F5). Takes approximately 2-3 minutes.
79. The script produces 6 result grids — each maps to a report table.

Important: The script runs DBCC DROPCLEANBUFFERS and DBCC FREEPROCCACHE to clear cache before query benchmarks. Do not run other queries during execution.

Step 11.2: Capture Environment Specs (Table 5.7 + 5.8)

80. Find Result Grid [0] — Hardware (CPU, RAM) and Software (SQL Server version, OS).
81. Screenshot both grids. Copy values into Table 5.7 and Table 5.8 in your report.

Table 5.7: Hardware Specification (Template)

Component	Google Cloud VM Specification
Cloud Provider	Google Cloud Platform (Compute Engine)
VM Type	e2-standard-4 (4 vCPUs, 2 cores, 16 GB RAM)
Processor (CPU)	Intel Broadwell (or equivalent) — 4 vCPUs, 2 cores
Memory (RAM)	16 GB
Storage	128 GB SSD persistent disk
Operating System	Windows Server 2025 Datacenter (x86_64, built 20260211)
Region / Zone	asia-southeast1-a (Singapore)

Figure 5.10: Benchmark Environment — Hardware and Software Specification (Google Cloud e2-standard-4)

Step 11.3: Capture ETL Load Times (Table 5.1)

82. Find Result Grid [1] — 'Consolidated Layer timing' showing BatchID, LoadType, Layer, Steps, Records, DurationSec.
83. Screenshot this grid — it maps directly to Table 5.1.

Table 5.1: ETL Pipeline Execution Time Benchmarks (Template)

Layer	Objects	Full Load (sec)	Incremental (sec)	Records (Full/Incr)
Staging	9	10.6	23.8	58,800 / 58,813
Bronze	23	43.3	39.4	184,068 / 50
Silver	6	6.3	6.2	2,000 / 57,174
Gold	11	27.1	24.5	91,738 / 90,721
TOTAL	**49**	91.6	97.7	336,606 / 206,758

Figure 5.11: ETL Pipeline Execution Time — Layer-by-Layer Duration for Full Load vs Incremental Load

Step 11.4: Capture Query Performance (Table 5.2)

84. Find Result Grid [2] — 'Table 5.2 Format: Gold vs Raw Vault'.
85. Shows: Complexity, QueryName, Gold Layer (ms), Raw Vault (ms), Vault/Gold Ratio, Winner.
86. Screenshot this grid.

Table 5.2: Query Performance Benchmarks (Template)

Complexity	Query Description	Gold (ms)	Vault (ms)	Joins	Speedup
Simple	Tax by Category	11.5	130.6	2 vs 7	11.4x
Medium	Top Taxpayers by Payment	31.3	80.5	4 vs 8	2.6x
Complex	Monthly Revenue Trend	64.6	151.9	6 vs 10	2.4x

Figure 5.12: Query Performance Comparison — Gold Star Schema vs Raw Vault by Complexity Level

Step 11.5: Capture Storage and Change Detection

87. Find Result Grid [3] — Database sizes (MB per database + table-level storage).
88. Find Result Grid [5] — SAT versioning (TotalRows, CurrentRows, HistoricRows) and DIM SCD2 (CurrentRows, ExpiredRows).
89. Screenshot both grids.

Figure 5.13: Storage Overhead Analysis — Database Size Distribution Across Medallion Architecture Layers

Figure 5.14: Historical Tracking Proof — SAT Versioning (Current vs Historic) and DIM SCD2 (Current vs Expired)

12. Scalability Analysis (Optional)

Repeat the full load pipeline with different data volumes to populate Table 5.3. For each scale point:

90. Modify 02_TransactionData.sql to set @TaxpayerCount = N.
91. Re-run 01_CreateDatabaseStructure.sql + 02_TransactionData.sql.
92. Run 00_CleanAll_FreshFullLoad.sql to clean the DW.
93. Execute Master_Complete_Pipeline (full load) via SSIS.
94. Run 13_Performance_Benchmarks.sql section [4] for throughput.
95. Record: Total Records, Duration (sec), Records/Second.

Table 5.3: ETL Pipeline Scalability Analysis (Template)

Taxpayers	~Records	Full Load (sec)	Records/sec	Linear Proj.
250	~15,000	[measure]	[calculate]	-
500	~30,000	[measure]	[calculate]	[project]
1,000 (base)	336,606	91.6	3,659	[project]
2,000	~120,000	[measure]	[calculate]	[project]
5,000	~300,000	[measure]	[calculate]	[project]

Note: Linear Projection = (Records at scale / Records at 250) x Time at 250. Comparing actual vs projected reveals linear, sub-linear, or super-linear scaling.

13. Screenshot Checklist

Figure	Caption	Report Section	Status
5.5	Clean Environment Verification — All DW Tables Reset to Zero Rows	Appendix	☐
5.6	SSIS Master Pipeline Full Load Execution — 49 Steps Completed	Section 4.5	☐
5.7	Full Load Row Count Verification — 49 Tables with Expected Counts	Section 5.1	☐
5.8	SSIS Incremental Load Execution — Watermark-Based Delta Path	Section 4.5	☐
5.9	Incremental Change Detection — Step Log with Delta + SAT Versioning	Section 5.2.2	☐
5.10	Benchmark Environment — Hardware and Software Specification	Table 5.7+5.8	☐
5.11	ETL Pipeline Execution Time — Layer-by-Layer Full vs Incremental	Table 5.1	☐
5.12	Query Performance — Gold Star Schema vs Raw Vault Comparison	Table 5.2	☐
5.13	Storage Overhead — Database Size Across Medallion Layers (MB)	Section 5.3	☐
5.14	Historical Tracking Proof — SAT Versioning + DIM SCD2 Metrics	Section 5.2.2	☐
5.15	ETL Throughput — Records/Second for Full and Incremental Loads	Section 5.3.1	☐
5.16	Scalability Analysis — Throughput at 250/500/1K/2K/5K Taxpayers (optional)	Table 5.3	☐

Appendix A: Complete Execution Checklist

#	Phase	Task	Time	Status
1	A	Create Google Cloud VM (e2-standard-4, Windows Server 2025)	5 min	☐
2	A	Set Windows password + Connect via RDP	5 min	☐
3	A	Install SQL Server 2025 Developer + SSMS + VS 2022 Community + SSIS	45 min	☐
4	A	Transfer files + Run SQL scripts 01-09	15 min	☐
5	A	Deploy SSIS project + Update connection strings + Test run	10 min	☐
6	A	Configure advisor access (RDP or SSMS)	5 min	☐
7	B	Clean environment (00_CleanAll) + Full Load + 10_Verify	3 min	☐
8	B	Insert test data (12_Incremental) + Incremental Load + 11_Verify	3 min	☐
9	B	Run 13_Performance_Benchmarks.sql + capture all result grids	5 min	☐
10	B	(Optional) Scalability runs at 250/500/2K/5K taxpayers	30 min	☐
11	All	Capture all 12 screenshots for report figures	15 min	☐
		TOTAL ESTIMATED TIME	**~2.5 hours**	

Tip: Complete Phase A once (setup is one-time). Phase B can be repeated anytime to re-capture benchmarks.

Appendix B: Troubleshooting

Problem	Cause	Solution
RDP connection refused	Firewall rule missing	VPC Network > Firewall > Verify default-allow-rdp exists for TCP 3389
SQL Server not starting	Service not running or paths wrong	Open SQL Server Configuration Manager > Verify service is running
SSIS connection error	Connection string not updated for VM	Update all 7 Connection Managers to localhost
Script timeout on 02_TransactionData	Large data generation takes time	Wait ~30-60 seconds; increase SSMS query timeout if needed
SSIS package fails with 'Login failed'	Connection uses wrong authentication	Set all CMs to Windows Authentication with localhost
VM still charging after stop	VM not stopped from Cloud Console	Always Stop from Google Cloud Console > Compute Engine to deallocate
Benchmark results inconsistent	Background processes or cache	Close all apps; run benchmark twice; use second run
LoadType column error in 11_Verify	Old script version	Use the updated 11_Verify_IncrementalLoad.sql (LoadType removed)